Life Cycle Analysis of Floating Offshore Wind Turbine Concepts

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1 Introduction

The International Energy Agency (IEA) Net Zero Emissions by 2030, which is a normative scenario showing a pathway for the world's energy industry to achieve net-zero CO_2 emissions by the year of 2050, meets the United Nation's Sustainable Development Goals (SDGs) [1]. According to the IEA, by 2021, of the total 830 GW wind capacity installed in the world, only 7% were offshore wind farms, while 93% were located onshore [2]. While onshore wind is a developed technology with a location in 115 countries, offshore wind is at a relatively early expansion stage with 19 countries currently having such systems. Offshore presence is however expected to increase in the upcoming years as more and more countries are in the developing stages of their first offshore wind farms. Windmills, used for grinding grain and pumping water, have a long history dating back to seventh-century Persia (Iran) [3]. However, generating electricity with wind is a much newer development. Large-scale wind turbines for electricity generation only emerged in the 1930s, primarily in

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Europe and the USA. There are several benefits of offshore wind power compared to onshore wind power. Firstly, offshore wind turbines can generate more electricity due to higher wind speeds and less turbulence over water. Secondly, offshore wind farms can be located closer to population centers, reducing transmission costs, and improving energy security. Thirdly, offshore wind turbines have less visual impact on the landscape and can be placed in deeper waters where they are less likely to interfere with shipping lanes or fishing grounds. The FOWT farms can create jobs in coastal communities and act as a reliable source of clean energy while minimizing environmental impacts.

Life Cycle Assessment (LCA) is a quintessential tool to assess the environmental impacts associated with all stages of a product's life cycle and helps to improve sustainability (see Curran [4]). Yildiz et al. [5] performed LCA of a Barge-Type Floating Wind Turbine and Comparison with Other Types of Wind Turbines. The authors made an environmental comparison of three scenarios of a 2 MW wind energy system—onshore, offshore floating, and jacket offshore wind (OW) turbines. The study revealed that the greatest environmental impact corresponds to the manufacturing stage of the system components steel and concrete emissions, and total GHG emissions stated in GWP for the offshore floating concept was 18.6 g CO_2-eq./kWh, the onshore concept had 7.09 g CO_2-eq./kWh, and the jacket offshore concept had 9.49 g CO_2-eq./kWh. Authors from the leading French OW development firm Ideol SA (now BW Ideol, Japanese) studied an offshore unit involving square ring floater. They have recommended that the construction could either be a steel-concrete combination or could be made purely of steel [6]. Calculated results were 3376 and 1468 tons (tot/8250 T = 0.409 t CO_2/T and 0.177 t CO_2/T) of CO_2 emissions for the concrete concept, and 7018 and 2420 tons (3.19 T and 1.1 T CO_2/T) of CO_2 emission for the steel concept—with new and recycled materials, respectively. The study found that concrete hulls are heavier but offer better stability, while steel hulls are lighter but require more maintenance, and that a steel hull's carbon content is twice as large as a concrete hull. The comparative study of concrete and steel substructures for FOWT was reported in DNV's project study [7] to investigate the possibility of establishing an industrial plant to produce concrete structures for the offshore wind industry. The objective of the study was the comparison of two offshore floating concepts made in steel or concrete, and results indicate the concrete concept has both lower financial costs and carbon emissions compared to steel. The study concludes that Norwegian firms are capable to deliver concrete concepts target of 67 units per year, while the steel industry is not capable of the same. Calculated results show that the carbon emission of concrete is 2.5–3 times lower than those of steel; even with even a 70% recycled steel used in the Chinese plants, concrete remains the more favorable. For the Norwegian National Framework for Wind power, the NVE performed a study on wind power emissions [8]. Their report provides an insight into LCA of wind turbines mounted on Norwegian land and compared wind power to other sources of energy. The report concluded with the fact that the leading environmental impact of these turbine systems came from the production of concrete for the foundation and transport of wind farm components. GWP values stated in the framework were that wind power has a GWP of 3–46 g

CO_2/kWh compared to gas- and coal energy sources with 500 and 1000 g CO_2/kWh, respectively. The life-cycle energy and environmental emissions of a typical offshore wind farm in China is reported by Yang et al. [9] where the authors have presented the emission from China's first offshore wind power farm. They performed an LCA study in which Life Cycle Inventory (LCI) assessment is reported to list the system components based on GHG emissions and reported a GWP emissions of 25.5 g CO_2-eq/kWh over the lifetime.

On March 29th, 2023, the Office of the Prime Minister, and OED—the Ministry of Petroleum, announced the first public tender for offshore wind projects in two areas of the North Sea—Utsira Nord and Sørlige Nordsjø II. The aim is to increase renewable energy production and reduce GHG emissions, a big step toward Norway's aim of 30 GW offshore wind by 2040. The government will propose a proposition to Parliament with a proposal for a cost framework and commitment authorization. The projects that receive support are awarded support for 500 MW each but can build up to 750 MW, while the project that does not receive state funding retains the right to the area and may use the public support system and participate in future competitions.

In the OW industry, both steel and concrete are critical, and while the sector has a low environmental impact compared to traditional fossil fuel energy sources [10], there are still carbon emissions to consider. The manufacturing process of the OW turbine structure requires significant amounts of energy and resources, including steel, concrete, surface treatment chemicals and the transportation to site. The extraction and processing of building materials and chemicals can result in GHG emissions. Transportation of the structures to the installation site also requires large vessels with emissions. The objective of this work is therefore to assess the carbon emission effects on the environment, comparing a hull made of steel vs. concrete, with the aim of filling the knowledge gap in LCA studies and raising carbon footprint awareness.

An industrial case scenario was used to conduct an LCA study, project layout projected (see Fig. 1), of the following: Case study scenario 1: steel hull, installed in Singapore, then shipped to Hanøytangen, Norway—Chinese steel origin (see Fig. 2a, b). Partial goal: determine EN-15804- and NORSOK-compliant surface coating system. Case study scenario 2: concrete hull, produced, installed, in periphery of Hanøytangen, Norway—Norwegian concrete origin (see Fig. 2c). Before gaining interest in this research study, a relevancy analysis was performed to use an industry specific case scenario applicable to the LCA study (see Fig. 1). An approach was made to map engineering consultancy firms in the offshore structural field, and a finalized reach out was done to SEVAN SSP—with whom a correspondence resulted in the study task, and the following project data and system description (see Table 1).

The article is organized as follows. A brief methodology is described in Sect. 2, with the subsection of steel surface coating, followed by Sect. 3 of results and discussions, with the subsection of a result sensitivity analysis. Finally, conclusions are drawn in Sect. 4.

Fig. 1 SEVAN SSP SWACH™ wind

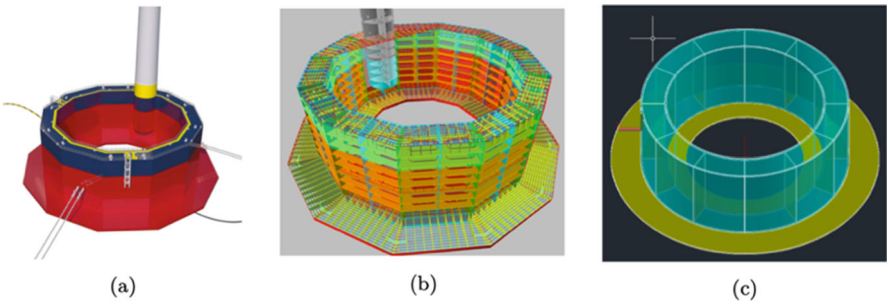


Fig. 2 SEVAN SSP-engineered FOWT concepts - a, b are steel, and c is concrete

Table 1 Project system description

Functional unit	Value	Description
Lifetime	50 years	The average technical lifetime of whole wind farm
No. of platform units	30	Study consists of 30 in steel versus 30 in concrete
Freight distance, SIN-OSL	10,000 nm	10,000 nautical miles equals 18,520 km
Freight distance, HAN-UTS	50 nm	50 nautical miles from Hanøytangen to Utsira

2 Methodology

In this section, we present the methodology for calculating CO_{2e}/T for steel vs. concrete FOWT units. It outlines the LCA analysis using chosen software and breaks down the steps of the study to find CO_{2e}/T, a material data assessment for

Fig. 3 Study-specific process flow chart

30 units steel vs. concrete, and GWP calculation in CO_{2e}/T. A flow chart (see Fig. 3) outlines the process. Iterations ensured compatibility with the SEVAN SSP case study.

As sustainability, climate change, and circular economy practices become more prevalent in the Engineering, Procurement, Construction and Installation (EPCI) industry, the demand for a time and cost-efficient method continues to grow. LCAs can be complex calculations, but there are digital tools and databases available to make the process easier. Ecoinvent is an online database that provides comprehensive information on the environmental impact of various products and services and contains data on the entire life cycle of a product, from raw material extraction to disposal. For this study the database search tool used was Ecoinvent's section for EN15804-compatible allocation and cut-off—since the system gives LCI describing waste and resource categories, excluding excessive by-products.

With so many LCA-software options on the market, it is important to determine a suitable one for our assessment. Tools considered were Ecochain Mobius, Ecochain Helix, GaBI, OneClickLCA, openLCA, and SimaPro, based on a selection criterion of having free trials, BIM compatibility, being web-based, while having a user-friendly Interface (UI). The appropriate software for this study was OneClickLCA, justified by the fact it has free online courses, help center with common FAQ, open-source learning material (YouTube, Vimeo), BIM-compatibility (IFC-file readability), and discounted academic and trial licenses.

OneClickLCA sources its data from a selection of sources, and they can be generic or product-specific if a producer has uploaded an Environmental Product Declaration (EPD) available for each country's EPD Program Operator and publisher. To shortly describe the software tool use process, a Life Cycle Inventory Assessment (LCIA) was done based on case study project specifications. Concrete concept hull design weighs 20,500 tons, of which 2000 tons are steel reinforcement bars—rebar/concrete ratio subject to confidential research. The 18,500 tons of concrete (high-strength C60/C70-grade NORSOK-compliant with 10900PSI compressive strength) was given A2 mode transport distance from producer to storage of 70 kilometers onboard a 40-ton trucks with EURO-6 engine with 100% capacity. Subsequently, A4 shows 26 kilometers transport distance from storage to project site.

2.1 Coating System

A partial objective is to determine an EN15804- and NORSOK M-501-system1-compliant surface treatment system for the steel hull. Method is to manually

Fig. 4 Coating system selection process chart

determine the case study needs (see Fig. 4), of a specific product and its feasible availability, GWP given in CO_{2e} per Ton, and total GWP for the 30-unit steel case scenario.

The NORSOK M-501 system 1 [11] is the requirement for the selection of a compliant coating system given by communication with engineers from SEVAN SSP—selected from the range of choice spanning from system no. 1 to system no. 9.

Selection criteria determined coating system no. 1, and to continue the determination of a product-specific EPD compliant with the PCR rules of EN15804, GWP stated in CO_{2e}, and a factory proximity to East-Asian Sea ports, a survey of available coating systems on the EPD Norway library was done. This resulted in the elimination of brands and factory locations from Norway, Germany, India and resulted in Jotun Jotamastic 70, Zhangjiagang, China, with EPD number: 1599-624-EN. The basis of the choice of the given factory location roots in the fact that there is a seaport proximity, for the easiness of shipping to the Singapore installation site.

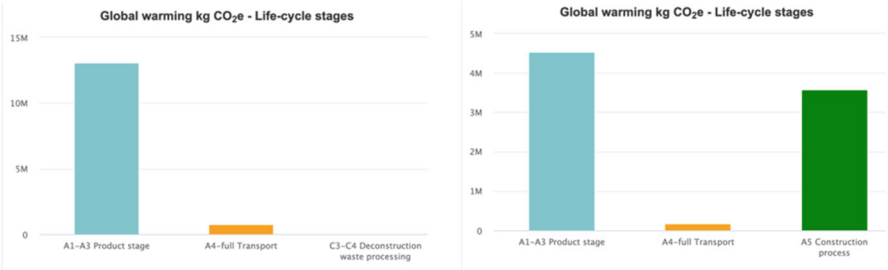
Results found in the EPD [11] with a EN15804-product stage system boundary A1-A3, *cradle-to-gate*, consist of A1: raw materials extraction, production, packaging, then A2: transport of raw materials and packaging, and finally A3: product stage: preassembly, pigments mixing dispersion, quality control, filling, and labeling. The calculations necessary to find GWP is based on; the EPD data of 3.19 kg CO_{2e} per kg paint, film thickness range 7.7–3.1 m^2/l, density (ρ) of 1.6 kg/l, with a project specific steel hull surface Area = 30,430 m^2, resulting in total masses of 6323.12 kg and 15705.81 kg paint, simply using $m = \rho/V$ for both film thicknesses.

3 Results and Discussions

Steel concept LCA study resulted in having the environmental impact parameter GWP of 13,821.257 T CO_{2e}/T, for the 5000-ton steel concept which is equivalent to the study selected unit of 2.76 T CO_{2e}/T (see Fig. 5a).

As for the NORSOK-compliant Jotun Jotamastic 70, the results are calculated to: Total GWP for spread rate 1 with 7.7 m^2/l = 30 × 6323.12 kg × 3.19 kg CO_{2e}/kg = 6051 T CO_{2e} Total GWP for spread rate 2 with 3.1 m^2/l = 30 × 15,705.81 kg × 3.19 kg CO_{2e}/kg = 1503.1 T CO_{2e}. The steel concept surface coating system necessary for 30 units has GWP ranging from 605.1 T to 1503.1 T CO_{2e}/T.

The concrete concept gives a GWP of 8298.743 T CO_{2e}, for the 20,500-ton concrete structure of which 2000 tons are reinforcement bars which is equivalent to the study selected unit of 0.404 T CO_{2e}/T (see Fig. 5b).



(a) LCA Results Steel units. (b) LCA Results Concrete units.

Fig. 5 (**a**, **b**) LCA results

Note that this is per system boundary of A1-A3. Installation, operation, maintenance stage is not defined in this case. The steel concept GWP per ton is 6–7 times higher, but that is only per weight. The project concept states 30 units, of which each concrete hull has 20,500 T while the steel concept only has 5000 T—equaling a 1/4 steel vs. concrete hull weight ratio which makes the steel concept only 1.77-times higher than concrete—equaling 70% worse than the concrete in total project mass.

Few LCA studies are performed on FOWT structures involving steel or concrete—results from this study are therefore relevant as supplementary scientific material. In relation to the aforementioned literature studies, the GWP calculation of this study with 0.404 T CO_{2e}/T for concrete and 2.76 T CO_{2e}/T for the steel concept falls within Choisnet et al.'s published article where the impacts amounted in 1.1–3.19 T CO_{2e}/T for steel and 0.177–0.41 for concrete. Simultaneously the NVE National Framework concluded with the fact that concrete production has a significant environmental impact, which confirms the graph from this study's result section where stage A1-A3 is shown in the results (see Fig. 5a, b).

Regarding the selection of the 10,900-PSI C60/C70 grade high-density concrete selected in OneClickLCA for the study, an estimated 10% recycled additive percentage was a composition of the cement used in the concrete mix—typical additives being hydraulic binders, such as fly ash, fume, and slag—all by-products from industrial processes. In case of a change in percentage of the reuse, a potential carbon emission reduction can occur. Typically, these additives also have their own GWP, not stated transparently although they are sources of CO_2 emissions.

Regarding the surface coating, read on page 8 of the source [11], the electricity mix in China is stated with a GHG of 1194.03 g CO_{2ekv}/kWh, by Ecoinvent. In China the GHG emissions from electricity is presumably higher than electricity mixes in Norway where 98% of energy sources are from renewable sources, while we in China have traditional fossil sources such as coal, heavy fuel, etc. An alternative solution is to treat the surface at dock in Hanøytangen. Note that this alternative must consider labor cost changes and shipping distance recalculations.

3.1 Sensitivity Analysis of the LCA Results

As the steel reinforcing bars of the concrete hull are a matter of early-stage research, even stated by DNV as a 2023-Q1-starting industrial collaborative research project, an awareness disclaimer must be made to the reader that the 9% steel content was used for calculation simplicity while DNV uses 2% [12]. As 8% of total emission in the steel concept LCA calculations represents the steel reinforcement bars, this is a source of emission which could possibly affect the total GWP if a higher recycling rate is used in its production.

As steel is the main material used to manufacture the 30-unit hulls, the upstream process is highly determinant of the total emission stated in the results. Even a low change in the steel recycling grade percentage during production in China could be significant to total emissions.

4 Conclusions

In today's world, amidst growing environmental concerns and resource scarcity, sustainability in engineering and energy efficiency has become more paramount than ever before. With the increasing demand for energy and the depletion of natural resources, it is crucial that we find ways to reduce our carbon footprint and preserve the environment for future generations.

Wind power is one of the most promising sources of renewable energy, and it has the potential to provide clean, reliable power and a significant portion of our global energy needs. Wind power is also a promising step toward reaching the UN SDGs, including SDG 7: Affordable and Clean Energy, SDG 9: Industry, Innovation and Infrastructure, and SDG 12: Responsible Consumption and Production. By investing in sustainable engineering practices and embracing energy efficiency, we can create a brighter future for ourselves and our planet. We have the power to make a difference, and by working together toward a common goal, we can create a world that is cleaner, healthier, and more sustainable for all.

An LCA study has been a great hands-on approach to achieving and comprehending the actual carbon footprint when choosing between concept 1 (steel) and concept 2 (concrete). This study has allowed us to compare the environmental impacts of the two materials over their life cycle, from the extraction of raw materials to the manufacturing, transportation, and installation, of the FOWT structures.

An environmental approach to industrial decision-making can make a difference and help the energy sector reach the targets of the UN SDGs. By considering the environmental impacts of our choices, we can select materials and processes that minimize our impact on the planet.

This work aims to share the knowledge gained during this research, hoping it proves beneficial for readers from both academic and industrial backgrounds. The aim was to make it comprehensible for the reader, by formulating it apprehensible without the reader being required to possess significant knowledge on LCA as a scientific field. As a conclusion, LCA studies can be a valuable tool for identifying opportunities to reduce the carbon footprint of FOWT structures.

4.1 Limitations and Future Work

As a disclaimer, it must be noted that challenges of the study were mainly limitations of material database availability and data variations on the A1-A3 analysis specter.

- Lack of sufficient EPD's from steel and concrete plant material databases.
- Limited data availability from the actual steel plants in China other than the single source used in this project in OneClickLCA.
- Average travel speed variations: shipping vessel carrying the two steel structures.
- Emission reduction of the steel industry will further reduce the GWP of FOWT farms. In the future, more data needs to be collected to improve LCA reliability.
- According to the NVE, in southern parts of China, the raw material stage's mining processes has affected loss of ecosystems and biodiversity [8].
- Jotun Jotamastic 70: the product service life is highly dependent on the use conditions; the GWP was calculated for a one-time application, and it depends on whether it needs reapplication or maintenance within the 50-year lifespan.

4.2 Declaration of Interests

As transparency in research is important, the signatory author has performed a review process of a number of questions on matter of interest conflicts— from a financial, professional, and personal point of view. Ethical approval is not applicable for this theoretical study, and no monetary funding or endowment was involved. The author hereby declares that there are no known interests or personal relationships that have occurred to in any way influence the angle, methodology, or results of this study.

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